Worksheet 3A: Quadratics — FOIL and Expanding

Part A: The FOIL Method — Basic Mechanics (12 problems)

FOIL stands for **F**irst, **O**uter, **I**nner, **L**ast — a method for multiplying two binomials.

Example: $(x+3)(x+2)$

```
First:  x · x = x²
Outer:  x · 2 = 2x
Inner:  3 · x = 3x
Last:   3 · 2 = 6

Combine: x² + 2x + 3x + 6 = x² + 5x + 6
```

Expand each expression completely. Show your F-O-I-L steps for the first few.

1. $(t+2)(t+5)$

2. $(p+6)(p+3)$

3. $(m+4)(m+7)$

4. $(r+1)(r+9)$

5. $(k+8)(k+2)$

6. $(n+5)(n+5)$ — *What do you notice about this one?*

7. $(w+3)(w+3)$

8. $(s+10)(s+1)$

9. $(h+6)(h+6)$

10. $(u+4)(u+4)$

11. What pattern do you see when both binomials are identical, like $(x+a)(x+a)$?

12. Complete this rule: $(x+a)^2 = x^2 + _____ + a^2$

Part B: WAIT — Exponents Don't Distribute Over Addition! (8 problems)

Common Mistake Alert!

Many students think $(a+b)^2 = a^2 + b^2$.

This is WRONG.

$(a+b)^2$ means $(a+b)(a+b)$, which you must FOIL out.

Let's prove this with actual numbers.

13. Let $a = 3$ and $b = 4$.

- (a) Calculate $(a+b)^2 = (3+4)^2 = 7^2 =$ _____
- (b) Calculate $a^2 + b^2 = 3^2 + 4^2 = 9 + 16 =$ _____
- (c) Are they equal? _____

14. Now let $a = 2$ and $b = 5$.

- (a) $(a+b)^2 =$ _____
- (b) $a^2 + b^2 =$ _____
- (c) What's the difference between the two answers? _____

15. Expand $(a+b)^2$ using FOIL:

$(a+b)^2 = (a+b)(a+b) =$ _____

16. The correct expansion is $a^2 + 2ab + b^2$. What's the "missing piece" when people incorrectly write $a^2 + b^2$?

17. Using $a = 3$ and $b = 4$ again: - (a) Calculate $2ab = 2(3)(4) =$ _____ - (b) Verify: $a^2+2ab+b^2 = 9 + 24 + 16 =$ _____ - (c) Does this match $(a+b)^2 = 49$? _____

18. Does the same mistake happen with subtraction? Test $(a-b)^2$ vs $a^2 - b^2$ with $a = 5$, $b = 2$: - (a) $(a-b)^2 = (5-2)^2 =$ _____ - (b) $a^2 - b^2 = 25 - 4 =$ _____ - (c) Are they equal? _____ - (d) What is the correct expansion of $(a-b)^2$? _____

19. Here's another trap. Is $(a+b)^3 = a^3 + b^3$? Test with $a = 2$, $b = 1$: - (a) $(a+b)^3 = (2+1)^3 = 3^3 =$ _____ - (b) $a^3 + b^3 = 8 + 1 =$ _____ - (c) Are they equal? _____

20. Complete the rule: **Exponents distribute over _____ but NOT over _____.**

(Hint: $(ab)^2 = a^2b^2$ works, but $(a+b)^2 \neq a^2 + b^2$)

Part C: FOIL with Negative Numbers (12 problems)

When negatives appear, keep careful track of signs.

Example: $(x-4)(x+3)$

```
First:  x · x = x²
Outer:  x · 3 = 3x
Inner:  -4 · x = -4x
Last:   -4 · 3 = -12

Combine: x² + 3x - 4x - 12 = x² - x - 12
```

Expand completely:

21. $(t-3)(t+5)$

22. $(p+4)(p-2)$

23. $(m-6)(m+1)$

24. $(r-2)(r-5)$

25. $(k-7)(k-3)$

26. $(n-4)(n-4)$ — *Squaring a binomial with subtraction*

27. $(w-8)(w-8)$

28. $(s+5)(s-5)$ — *What happens to the middle term?*

29. $(h+9)(h-9)$

30. $(u+7)(u-7)$

31. What pattern do you see when the binomials are $(x+a)(x-a)$?

32. Complete this rule: $(x+a)(x-a) =$ _____

Part D: FOIL with Coefficients (10 problems)

When the variable has a coefficient, apply FOIL the same way.

Example: $(2x+3)(x+4)$

```
First:  2x · x = 2x²
Outer:  2x · 4 = 8x
Inner:  3 · x = 3x
Last:   3 · 4 = 12

Combine: 2x² + 8x + 3x + 12 = 2x² + 11x + 12
```

Expand completely:

33. $(2t+1)(t+3)$

34. $(3p+2)(p+4)$

35. $(2m-3)(m+5)$

36. $(4r+1)(r-2)$

37. $(3k-2)(2k+1)$

38. $(5n+2)(2n-3)$

39. $(2w+3)(2w+3)$ — *Squaring with coefficients*

40. $(3s-1)(3s-1)$

41. $(4h+5)(4h-5)$ — *Difference of squares with coefficients*

42. $(2u-7)(2u+7)$

Part E: Patterns Worth Memorizing (8 problems)

These patterns appear so often that recognizing them instantly saves time.

Pattern 1: Perfect Square Trinomials

$$\boxed{(a+b)^2 = a^2 + 2ab + b^2}$$

$$\boxed{(a-b)^2 = a^2 - 2ab + b^2}$$

*The middle term is always **twice** the product of the two terms.*

Pattern 2: Difference of Squares

$$\boxed{(a+b)(a-b) = a^2 - b^2}$$

The middle terms cancel out, leaving only the difference of squares.

Use the patterns to expand quickly:

43. $(t+8)^2$

44. $(p-5)^2$

45. $(m+11)(m-11)$

46. $(3r+4)^2$

47. $(5k-2)^2$

48. $(6n+1)(6n-1)$

49. $(2w+9)^2$

50. $(4s-3)(4s+3)$

Part F: Discovering Pascal's Triangle (12 problems)

Let's expand powers of $(a+b)$ and look for patterns in the coefficients.

51. $(a+b)^0 = 1$

Coefficients: **1**

52. $(a+b)^1 = a+b$

Coefficients: **1, 1**

53. Expand $(a+b)^2$ using FOIL.

$(a+b)^2 =$ _____

Coefficients of a^2, ab, and b^2: _____, _____, _____

54. Expand $(a+b)^3$ by computing $(a+b)\cdot(a+b)^2$.

Use your answer from problem 53 and distribute $(a+b)$ through it.

$(a+b)^3 =$ _____

Coefficients: _____, _____, _____, _____

55. Expand $(a+b)^4$ by computing $(a+b)\cdot(a+b)^3$.

$(a+b)^4 =$ _____

Coefficients: _____, _____, _____, _____, _____

56. Arrange your coefficients in a triangle:

```
          1                    (a + b)
        1   1                  (a + b)¹
      1   2   1                (a + b)²
    1   3   3   1              (a + b)³
  1   ?   ?   ?   1            (a + b)
```

Fill in the missing numbers for row 4.

57. Look at how each number relates to the two numbers above it. Describe the pattern:

_____

58. Using the pattern, predict row 5 (the coefficients for $(a+b)^5$):

_____, _____, _____, _____, _____, _____

59. This triangle is called _____**'s Triangle**.

(Named after a 17th-century French mathematician, though mathematicians in Persia, China, and India discovered it centuries earlier.)

60. Expand $(a+b)^5$ using your predicted coefficients:

$(a+b)^5 =$ _____

61. Bonus connection: The number in row n, position k tells you "how many ways can you choose k items from n items?"

Row 4, position 2 has the number **6**.

Verify: List all ways to choose 2 letters from {A, B, C, D}:

_____

(There should be exactly 6 ways!)

62. Using the triangle, how many ways can you choose 3 items from 5 items?

(Look at row 5, position 3)

_____

Part G: Beyond Two Terms (8 problems)

For multiplying larger expressions, we distribute systematically.

Example: $(x+2)(x^2+3x+1)$

```
Distribute x:   x·x² + x·3x + x·1    = x³ + 3x² + x
Distribute 2:   2·x² + 2·3x + 2·1    = 2x² + 6x + 2
Combine:        x³ + 3x² + 2x² + x + 6x + 2 = x³ + 5x² + 7x + 2
```

Expand completely:

63. $(t+3)(t^2+2t+4)$

64. $(p-2)(p^2+4p+1)$

65. $(m+1)(m^2-m+1)$ — *Something interesting happens here!*

66. $(r-1)(r^2+r+1)$ — *And here!*

67. $(2k+1)(k^2-3k+2)$

68. $(n+2)(n+3)(n+1)$ — *Do two at a time*

69. Expand $(w-1)^3$ by first finding $(w-1)^2$, then multiplying by $(w-1)$.

70. Expand $(2t+1)^3$ using the same method.